

SONiX 32-Bit Cortex-M0 Micro-Controller

WDT Example Code

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AMENDMENT HISTORY

Version	Date	Description
1.0	2024/12/02	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of WDT.

2 FEATURE

- Reset Mode : Generate a reset to reset system every 250ms.
- Interrupt Mode : Generate an interrupt every 250ms.

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Reset Mode

1. Open WDT.h, and set WDT_MOD as RESET.

```
7  /*_____ D E F I N I T I O N S _____*/
8  #define RESET                0                //WDT as Reset mode
9  #define INTERRUPT            1                //WDT as Interrupt mode
10 #define WDT_MODE              RESET
```

2. Call WDT_Init();

```
int main (void)
{
    //init WDT
    WDT_Init();

    .
    .
    User code
    .
    .
}
```

3. Modify the settings inside the following red circle to set the WDT clock source and WDT reload value.

Example: Set WDT clock source as WDT_CLKSEL_ILRC

- 1) For SN32F700(B)/100/770 (ILRC=16KHz)
 → WDT reload value = 31
- 2) For SN32F760(B)/240(B)/260/780/280 (ILRC=32KHz)
 → WDT reload value = 61

```
void WDT_Init(void)
{
    uint32_t wRegBuf;

    WDT_SelectClockSource(WDT_CLKSEL_ILRC);           //clock source select

    #if WDT_MODE == INTERRUPT
        wRegBuf = mskWDT_WDTIE_ENABLE;               //WDT as interrupt mode
    #endif

    #if WDT_MODE == RESET
        wRegBuf = mskWDT_WDTIE_DISABLE;              //WDT as reset mode
    #endif

    wRegBuf = wRegBuf & (~mskWDT_WDTINT);             //Clear WDT interrupt flag

    wRegBuf = wRegBuf | mskWDT_WDKEY;

    SN_WDT->CFG = wRegBuf;

    WDT_Reload_Value(61);                            //Set overflow time = 250ms

    WDT_NvicEnable();                                 //Enable WDT NVIC interrupt

    __WDT_ENABLE;                                    //Enable WDT
}
```

4.2 Interrupt Mode

1. Open WDT.h, and set WDT_MOD as INTERRUPT.

```
7  /* _____ D E F I N I T I O N S _____ */
8  #define RESET                0                //WDT as Reset mode
9  #define INTERRUPT            1                //WDT as Interrupt mode
10 #define WDT_MODE              INTERRUPT
```

2. Call WDT_Init();

```
int main (void)
{
    //init WDT
    WDT_Init();

    .
    .
    User code
    .
    .
}
```

3. Modify the settings inside the following red circle to set the WDT clock source and WDT reload value.

Example : Set WDT clock source as WDT_CLKSEL_ILRC

1) ILRC=16KHz

→ WDT reload value = 31

2) ILRC=32KHz

→ WDT reload value = 61

```
void WDT_Init(void)
{
    uint32_t wRegBuf;

    WDT_SelectClockSource(WDT_CLKSEL_ILRC);           //clock source select

    #if WDT_MODE == INTERRUPT
        wRegBuf = mskWDT_WDTIE_ENABLE;               //WDT as interrupt mode
    #endif

    #if WDT_MODE == RESET
        wRegBuf = mskWDT_WDTIE_DISABLE;              //WDT as reset mode
    #endif

    wRegBuf = wRegBuf & (~mskWDT_WDTINT);             //Clear WDT interrupt flag

    wRegBuf = wRegBuf | mskWDT_WDKEY;

    SN_WDT->CFG = wRegBuf;

    WDT_Reload_Value(61);                             //Set overflow time = 250ms

    WDT_NvicEnable();                                  //Enable WDT NVIC interrupt

    __WDT_ENABLE;                                     //Enable WDT
}
```

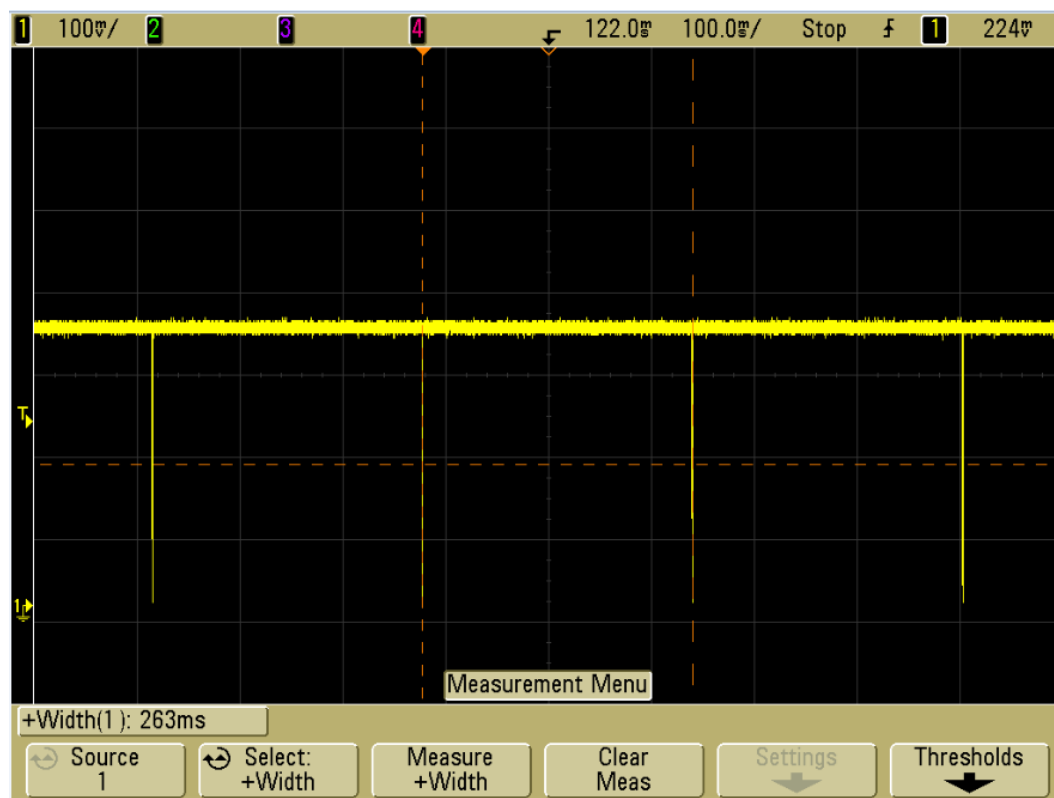
5 RESULT

5.1 Reset Mode

The WDT is configured to generate a reset to reset system every 250ms.

The output can be observed from the GPIO P2.0WDT start counting to generate the reset time, results in the following figure.

* Note: ILRC may have some deviation (+/-20%). So, actual reset time may have some deviation (+/-20%).

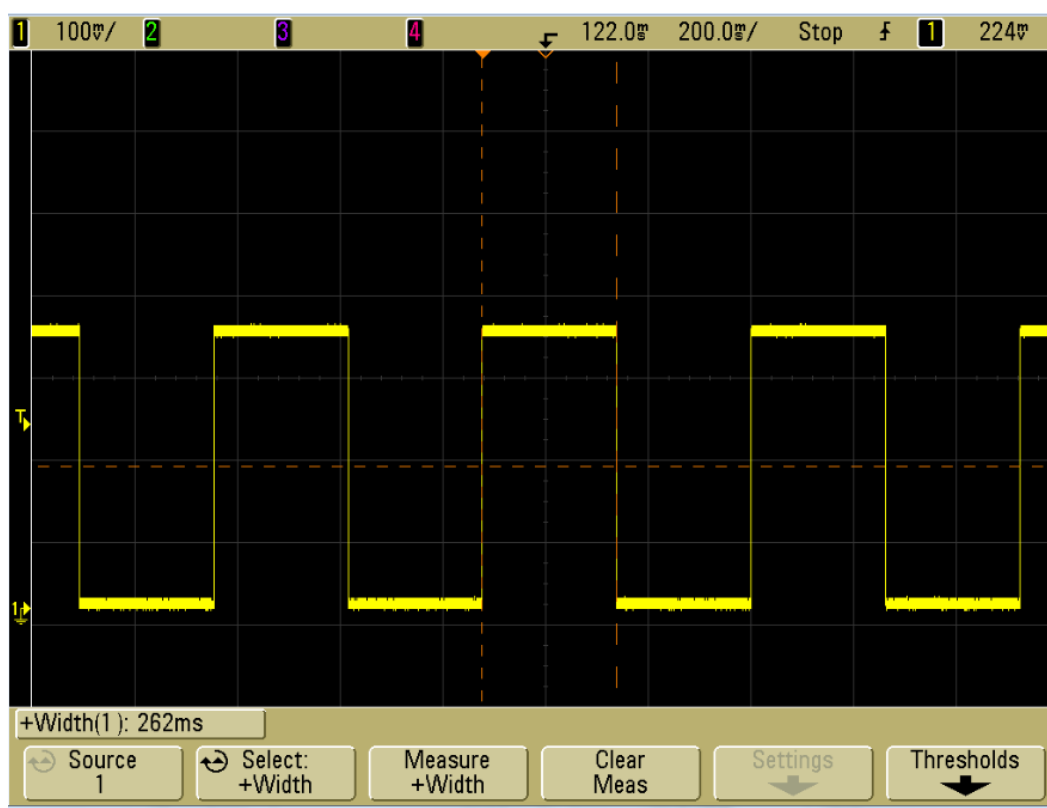


5.2 Interrupt Mode

The WDT is configured to generate an interrupt every 250ms.

The output can be observed by the GPIO P2.0 toggle every 250ms, results in the following figure.

* Note: ILRC may have some deviation (+/-20%). So, actual reset time may have some deviation (+/-20%).



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